

FRANK DITZ

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SUMMARY

Applied AI software engineer building enterprise LLM applications and agentic tool workflows that connect natural-language requests to governed data systems. Promoted from intern to associate while delivering software and secure integrations for 3 enterprise customers across Snowflake, PostgreSQL, Alation, and AWS.

EXPERIENCE

ALTR

Associate Field Software Engineer - Data Governance & Security
Field Engineering Intern

Melbourne, FL
Sep 2025 - Feb 2026
Aug 2024 - Aug 2025

- Promoted from intern to associate while delivering software and secure integrations for 3 enterprise customers and building applied AI workflows for enterprise data access.
- Designed Python, Node.js, SQL, and REST API workflows connecting LLM tools to metadata catalogs, databases, and cloud services with authentication, validation, and explicit failure handling.
- Built and validated governed data-access patterns using RBAC, masking, tokenization, encryption, and query safeguards so LLM workflows preserved existing enterprise controls.
- Diagnosed defects across session state, callbacks, authentication, database roles, and cloud services using targeted tests, structured logging, and direct workflow validation.
- Automated encryption of newly ingested Fivetran data in Snowflake using sync metadata and idempotent processing, reducing the window in which raw PII remained exposed.

Ditz Medical & Laser Aesthetics Center
Manager

Melbourne, FL
Jan 2020 - Jul 2024

- Managed daily operations across scheduling, billing, inventory, staff coordination, and client communication.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript

Applied AI: OpenAI API, LLM applications, tool calling, agent workflows, prompt engineering, structured outputs, metadata retrieval

Backend & Integration: FastAPI, Node.js, REST APIs, SDK development, Pydantic, authentication

Data & Cloud: Snowflake, PostgreSQL, Alation, Streamlit, AWS Lambda, Step Functions, S3, Docker, Terraform

Security & Reliability: RBAC, masking, tokenization, encryption, pytest, MyPy, Ruff, logging, retries, pagination, GitLab CI/CD

SELECTED APPLIED AI & ENGINEERING PROJECTS

AI-Powered Enterprise Data Assistant

- Built a Python/Streamlit application using OpenAI tool calling across Snowflake, PostgreSQL, and Alation, enabling non-technical users to ask natural-language questions and receive governed answers, tables, and charts.
- Designed metadata-first tools for catalog lookup, schema inspection, query execution, and response formatting, grounding source selection in Alation metadata before database execution.
- Implemented Snowflake RSA key-pair authentication, environment-based secrets, RBAC, and query safeguards; resolved session state, streaming responses, callbacks, and connector reliability issues.

Python SDK for ALTR APIs

- Designed and built a typed Python SDK wrapping 100+ REST API endpoints into a reusable client library with a shared session layer, modular resource mixins, Pydantic models, authentication, pagination, retries, structured exceptions, and logging.
- Established GitLab CI with Ruff, MyPy, and pytest to enforce code quality, type safety, and regression checks for a library other engineers could maintain and extend.

Pre-Ingestion PII Tokenization Pipeline

- Built a serverless AWS-to-Snowflake workflow with checkpoint recovery, pagination, retries, persistent state, audit logging, and record-level validation to process tens of thousands of rows reliably within Lambda limits.

EDUCATION

University of Central Florida - Data Analytics Bootcamp | Certificate

Aug 2023

University of Florida - Bachelor of Arts in Biology | Academic Scholarship

Dec 2018